Cross-Signer Generalization for Isolated ASL Word Classification on WLASL

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1. Problem Statement

Automatic sign language recognition (SLR) on WLASL often reports strong results, around 60–80% for some pose-based methods, but these numbers are normally obtained with signer-overlapping splits. In that setting, the same signer can appear in both train and test, so the model may learn part of *who* is signing instead of only *what* gloss is performed.

This project focuses on cross-signer generalization, one of the open challenges in Domain C of the brief. Zhou et al. (2023) describe subject transfer as still difficult for skeleton-based action models, especially when different people have similar appearance but different motion habits. AUTSL also shows the size of the problem: signer-dependent accuracy reaches 95.95%, while user-independent accuracy drops to 62.02% (Sincan & Keles, 2020). WLASL does not provide this protocol directly; in NSLT-100, 52 of 97 signers overlap between the official train and test splits.

Research question: Can a pose-based action recognition pipeline address cross-signer generalization for isolated ASL word classification on WLASL RGB sign clips when it is evaluated on a signer-disjoint held-out test set?

Initial hypothesis: Partially yes. Models that encode temporal joint motion rather than pixel appearance should show a smaller performance drop on unseen signers, because signer-specific variation hand morphology, skin tone, clothing, recording background is expressed mainly in the RGB signal while gloss identity is carried by articulation trajectories. Discarding pixels at the input stage and normalizing skeleton sequences into a linguistically motivated signing space should suppress the spurious appearance correlations that drive failure under signer shift. This prediction is grounded in Zhou et al. (2023, §6.2–6.3) and the signed-language normalization rationale of Boháček & Hrúz (2022).

2. Related Work

Boháček & Hrúz (2022) introduce SPOTER, a Transformer classifier for word-level sign recognition. It works on 54-joint 2D skeletons and applies signing-space normalization plus per-hand normalization. This is relevant here because the method removes RGB appearance before classification. Boháček et al. (2022) compare SPOTER with I3D and TimeSformer on WLASL and AUTSL, showing that pose-based processing can reduce dependence on background features. Dafnis et al. (2022) use graph-convolutional skeleton models, and Park et al. (2024) improve preprocessing with neck–palm anchor normalization and bilinear hand-keypoint reconstruction to reduce MediaPipe dropout.

RGB baselines such as I3D on WLASL (Li et al., 2020) and later training-strategy work (Li et al., 2024) are important references, but they do not evaluate signer-disjoint generalization. Continuous SLR models such as Haque et al. (2025) and Tran et al. (2025) do use signer-independent settings, but they solve sentence-level sequence decoding with WER, not isolated single-gloss classification.

The remaining gap is therefore specific: WLASL-100 has many reported results on signer-overlapping splits, but it is still unclear how far skeleton normalization alone can go when the test signers are completely unseen.

3. Architecture Selection

The architecture selection is guided by one criterion: which design properties suppress signer-specific appearance without discarding gloss-discriminative motion signal?

3.1 Chosen pipeline

We select a three-stage pose-based pipeline: **MediaPipe Holistic** (pretrained, inference only) for landmark extraction → SPOTER signing-space and per-hand bounding-box normalization → **SPOTER** Transformer classifier (Boháček & Hrúz, 2022; Apache-2.0, available at github.com/matyasbohacek/spoter). An optional fourth stage - Park et al. (2024) anchor normalization and bilinear hand reconstruction - is evaluated as the extension experiment.

Each design property targets the cross-signer challenge:

- **Skeleton-only input.** After MediaPipe extraction, all RGB channels are discarded. The classifier has no access to skin tone, sleeve color, background, or compression artifacts. This removes the primary channel through which signer identity leaks into training, implementing the recommendation in Zhou et al. (2023, §6.3.2) that skeleton inputs are more robust to appearance variation.
- **Signing-space normalization.** SPOTER projects body landmarks into a linguistically motivated signing box defined in head units. Camera distance, body scale, and frame position - all of which differ across signers and recording setups - are normalized out. Boháček & Hrúz (2022, §4.3) argue this step is necessary for spatial generalization.
- **Per-hand bounding-box normalization.** Each hand's landmarks are normalized within their own bounding box, so the model learns articulation geometry (relative joint distances and angles) rather than absolute screen position or hand size, reducing confounding between signer hand morphology and gloss identity.
- **Transformer temporal modeling.** Self-attention over the full 50-frame normalized pose sequence captures gloss-specific articulation timing across variable-length clips without a fixed short window - unlike clip-based 3D CNNs.

3.2 Alternatives considered and rejected

I3D RGB 3D CNN (Li et al., 2020). I3D fine-tunes Kinetics-pretrained 3D convolutions on RGB clip tensors. Three-dimensional convolutions integrate pixel appearance with motion at every layer; there is no structural mechanism to prevent the network from learning that a particular signer's hand texture or backdrop predicts a gloss. Under signer-disjoint evaluation this is exactly the failure mode being measured, not an incidental weakness. I3D was rejected as the primary architecture because it embodies the appearance-dominated pathway the hypothesis predicts will fail - not because it underperforms newer RGB models on signer-overlapping leaderboards.

TimeSformer (Boháček et al., 2022). Factorized space-time self-attention on RGB patch tokens can up-weight hand regions adaptively, but tokens still encode signer-specific appearance; attention re-weights rather than discards it. Quadratic memory over variable-length clips also makes fine-tuning expensive on the roughly 1,000-clip Kaggle corpus. The mechanistic problem is the same as I3D.

Signer-Invariant Conformer (Haque et al., 2025). This Conformer architecture targets continuous, sentence-level decoding with sequence alignment and CTC-based WER objectives. WLASL `nslt_100` provides pre-segmented, single-gloss clips; the sentence-decoding machinery addresses a problem we do not have. Rejection is on task-formulation grounds, not on signer-independence grounds.

The SPOTER pipeline was not chosen because it holds the top accuracy on signer-overlapping WLASL-100 rankings—it does not. It was chosen because its design systematically removes the appearance signal that drives signer-specific overfitting, making it the appropriate architecture to test the motion-centric generalization hypothesis.

4. Experimental Setup

4.1 Dataset

We use **WLASL v0.3**, restricted to the NSLT 100-class subset (`nslt_100.json`). The subset contains 2,038 annotated clips across 100 isolated ASL glosses and 97 signers. After filtering to the locally available videos, the working corpus contains 1,013 clips from 64 signers. Clips are trimmed using the NSLT `[start_frame, end_frame]` bounds. Class balance is mild, with 5–16 clips per gloss, but signer contribution is skewed, with 1–164 clips per signer. For that reason, we report macro-F1 together with top-1 accuracy.

4.2 Signer-disjoint evaluation protocol

The official NSLT train/val/test subsets are unsuitable for cross-signer evaluation: in `nslt_100`, 52 of 97 signers appear in both train and test. We constructed a **custom signer-disjoint split** by assigning each signer's entire set of clips to exactly one partition. Signers were sorted by ID and distributed round-robin in a 7:1:2 ratio, then a repair step reassigned individual signers to ensure all 100 classes appear in both train and test. No signer appears in more than one partition.

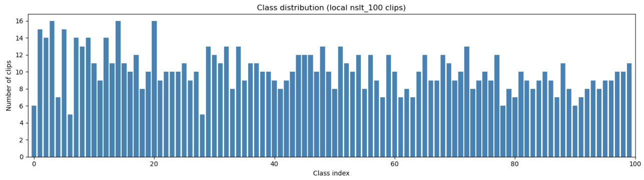

Figure 1: Class Distribution (Local NSLT-100 Clips)

Split	Signers	Clips
Train	45	721
Val	5	14
Test	14	278

Split metadata is stored at `dataset/splits/signer_disjoint_nslt100.json`. Validation metrics are used solely for model selection, all primary results are reported on the held-out test partition only.

4.3 Preprocessing, model, and metrics

Frames are extracted with OpenCV from the trimmed range. MediaPipe Holistic runs on every frame and produces body and hand landmarks; the neck is estimated as the midpoint between the shoulders. Each sequence is exported to SPOTER CSV format with 54 joints and 108 coordinates per frame, then padded or truncated to **50 frames**. SPOTER normalization is applied in the dataset loader. The Park extension uses the same baseline pose CSVs and applies anchor translation plus bilinear hand-keypoint reconstruction before SPOTER normalization.

SPOTER is trained from random initialization with `num_classes=100` and `hidden_dim=108`. Training uses SGD with lr 0.001, batch size 1, up to 100 epochs, ReduceLROnPlateau, Gaussian noise augmentation ($\sigma = 0.001$), built-in pose augmentations, early stopping on validation top-1 accuracy, and seed 379. The extension uses the same hyperparameters.

The two metrics are **top-1 accuracy** and **macro-F1**. Chance level is 1.00% for 100 classes. A majority-class baseline reaches 0.36% top-1 on the same test set.

5. Results

5.1 Main results

Setting	Test top-1	Test macro-F1
Majority-class baseline	0.36%	∼0.01%
Chance (1/100)	1.00%	1.00%
SPOTER (baseline poses)	**10.43%**	**7.8%**
SPOTER + Park normalization	**10.43**	**7.8%**

SPOTER is clearly above chance and the majority baseline, so normalized skeleton sequences contain useful gloss information. At the same time, the result is far below the usual WLASL-100 pose-based figures reported on signer-overlapping splits. This supports the idea that the official split hides part of the cross-signer difficulty.

Validation top-1 at the best checkpoint is 14.29% on 14 clips. Training accuracy reaches about 32% near early stopping, which already shows a large train–test gap.

Figure 2 description: Heatmap of all 278 signer-disjoint test predictions across 100 gloss classes. Off-diagonal mass is widespread, reflecting that errors are distributed across many gloss pairs rather than concentrated in a single class - consistent with a cross-signer generalization failure rather than isolated class difficulty.

5.2 Training dynamics and extension

The main pattern is overfitting to training signers: about 32% train accuracy against 10.43% test accuracy. Validation accuracy is close to the test result but unstable because the validation set has only 14 clips. The best checkpoint is therefore selected before the model fits too much to signer-specific motion patterns.

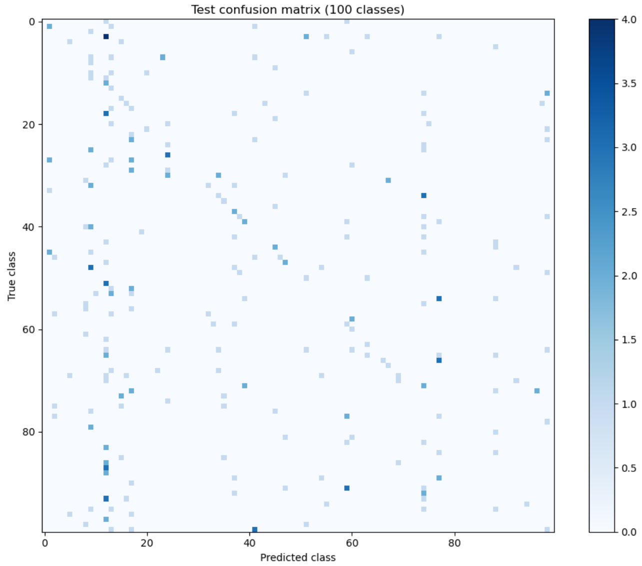

Figure 2: Test Confusion Matrix (100 Classes)

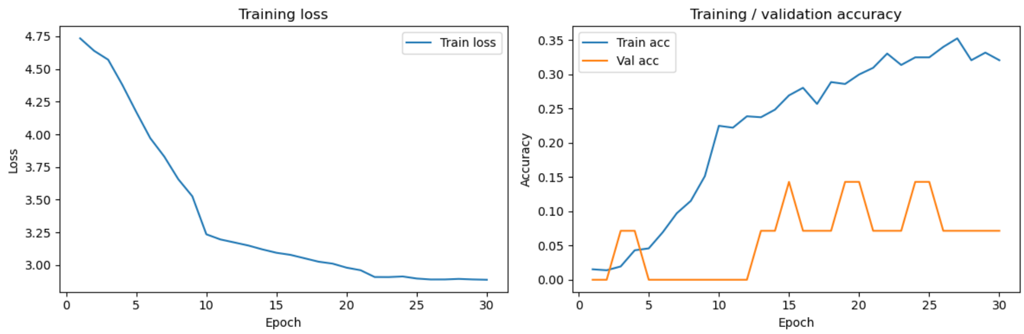

Figure 3: Training Loss and Accuracy Curves

Figure 3 description: Left panel: training loss over epochs. Right panel: training and validation top-1 accuracy. Training accuracy rises to approximately 32% while validation accuracy plateaus near 14%, visualizing the signer-specific overfitting that drives the train–test generalization gap.

The Park preprocessing extension gives the same result as the baseline: 10.43% top-1 and 7.39% macro-F1. This negative result is useful because it shows that anchor translation and 2D hand-keypoint reconstruction do not solve the main signer-disjoint error in this setting.

6. Failure Analysis

6.1 Failure 1. Cross-signer kinematic shift

Video / signer	57291, signer 12 (held-out test)
True gloss → predicted	*tell* (class 98) → *candy* (class 8)
Per-signer error rate	72 / 83 clips misclassified (86.7%)
Class *tell* recall	0.00%
Overall test top-1	10.43%

Signer 12 contributes 83 of 278 test clips and has an 86.7% error rate, so this is not just one isolated mistake. SPOTER normalization removes scale and frame position, but it does not remove signing speed, range of motion, rhythm, or MediaPipe estimation bias for a particular person. These signer-specific kinematics remain in the coordinates. The Transformer then maps the shifted trajectory to the nearest pattern seen during training, which causes many wrong gloss predictions for the same held-out signer.

Figure 4 description: Side-by-side RGB frame and MediaPipe skeleton overlay for a held-out test clip (true gloss: *tell*, predicted: *candy*). The skeleton is present and normalized, yet the model misclassifies - illustrating that coordinate normalization does not remove signer-specific kinematic habits.

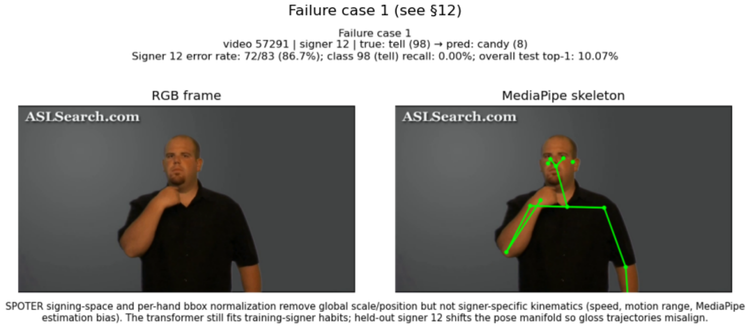

Figure 4: Failure Case 1 - Cross-Signer Kinematic Shift (Video 57291, Signer 12)

6.2 Failure 2. Fine-grained 2D pose collapse and hand dropout

Video / signer	05730, signer 32 (held-out test)
True gloss → predicted	*before* (class 3) → *help* (class 12)
Confusion count	4 before→help errors on test (top confused pair)
Test macro-F1	7.39%

This error is connected to the 2D input representation. MediaPipe Holistic gives 2D landmarks, so depth, grip contact, and some finger-closure information are lost. *Before* and *help* share partially overlapping 2D hand trajectories; under fast motion, hand landmarks may also be dropped or smoothed, making the SPOTER input vectors even more similar. Park-style interpolation can fill missing 2D points, but it cannot recover true depth or contact cues.

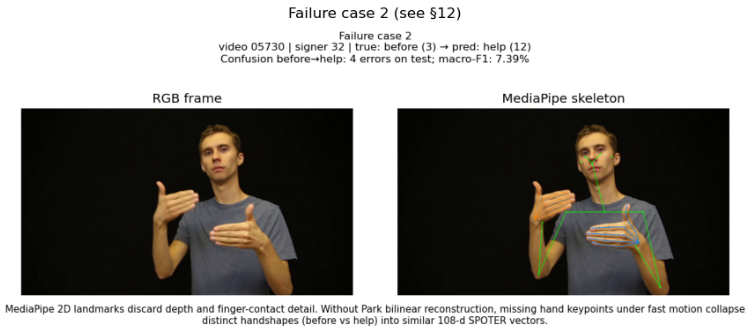

Figure 5: Failure Case 2 - Before→Help Confusion (Video 05730, Signer 32)

Figure 5 description: Side-by-side RGB frame and skeleton overlay for the top confused gloss pair on the test set (*before* misclassified as *help*). Similar 2D hand trajectories under MediaPipe dropout collapse distinct handshapes into near-identical pose vectors.

7. Thesis Direction

Did the results support the initial hypothesis?

Partially refuted. The hypothesis predicted that a skeleton-only pipeline with signing-space normalization would retain gloss-discriminative motion information while suppressing signer-specific appearance cues. The signer-disjoint experiment partially supports this hypothesis: SPOTER achieves 10.43% top-1 and 7.8% macro-F1 on 14 unseen test signers, clearly above chance (1%) and the majority baseline (0.36%). This indicates that normalized pose sequences contain transferable information about ASL gloss identity. However, the low absolute accuracy also shows that this representation is not sufficiently signer-invariant on its own.

The refuting evidence is the 32% train versus 10% test gap: normalizing coordinates in a head-unit signing box does not produce signer-invariant pose representations. The Transformer still learns motion patterns specific to the training performers. Applying a stronger normalization (Park et al., 2024 anchor preprocessing) produced no measurable improvement. The hypothesis that skeleton normalization is sufficient to close the signer-disjoint gap is not supported at this data scale and with this normalization scheme.

The result is informative rather than simply negative. It establishes that the gap between signer-dependent and signer-independent evaluation on WLASL-100 exists and is substantial (~22 percentage points relative to comparable training-set performance), replicating the pattern seen in AUTSL (Sincan & Keles, 2020), and confirms that RGB removal is necessary but not sufficient.

What the failure analysis reveals

Failure case 1 shows that signer-specific kinematics - not just appearance - survive SPOTER normalization and drive most of the test error. The architecture assumes that signing-space normalization removes the confounding variables; the results falsify this for kinematic habits. Failure case 2 shows that 2D monocular pose estimation loses depth and contact cues that are necessary to discriminate sign pairs with similar 2D projections, and that post-hoc keypoint reconstruction from sparse 2D data cannot recover what was never observed. These two failure modes operate at different levels: one is a normalization coverage problem, the other is a sensor modality problem.

Concrete next step

The dominant bottleneck is cross-signer kinematic shift. A feasible next step is signer-aware pose augmentation during training: temporally scale pose sequences (speed up / slow down) and add per-joint noise calibrated to the variance of held-out signer kinematics estimated from a small unlabeled sample. This approach requires no new labels, no additional video capture, and is directly applicable to the existing signer-disjoint split. It targets the distribution mismatch between training-signer and test-signer pose trajectories without changing the architecture or modality.

For failure case 2, the appropriate fix is richer input - either a depth camera or a lifted 3D pose from a multi-view or depth-aware pose estimator - rather than refinements to 2D coordinate coverage. This would require re-extracting poses but not re-annotating the dataset. Both directions are feasible on the WLASL-100 subset and can be evaluated with the existing signer-disjoint protocol.

LLM Assistance

LLMs were used for code refactoring and grammatical editing of the report. All research decisions, experiments, analysis, and conclusions were performed by the authors.

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